

## OFFICIAL COURSE SYLLABUS HOMEWORK · DENNIS G. ZILL (7TH EDITION)

## Chapter 1: Assigned Textbook Problems &amp; Step-by-Step Solutions

Comprehensive step-by-step engineering derivations, method classifications, and official textbook answer verification for all assigned problems in Section 1.1, Section 1.2, and Section 1.3.

25 Assigned Problems Total

100% Verified with Zill Solution Manual

Dual Edition Cross-Referenced (7th vs 6th)

Midterm 1 Preparation

## SECTION 1.1: DEFINITIONS AND TERMINOLOGY (PROBLEMS 1, 3, 5, 6)

Order &amp; Linearity Classification

## Problem 1.1.1 (Zill 7th Ed Section 1.1 #1 · 6th Ed #1)

ORDER &amp; LINEARITY

**QUESTION:** State the order of the given ODE. Determine whether the equation is linear or nonlinear:

$$(1-x)y'' - 4xy' + 5y = \cos x$$

**Step 1 (Find Order):** The highest-order derivative appearing in the equation is the second derivative  $y'' = \frac{d^2y}{dx^2}$ . Therefore, the ODE has **order 2**.

**Step 2 (Test Linearity):** An  $n$ -th order ODE is linear if it can be written as  $a_n(x)y^{(n)} + \dots + a_1(x)y' + a_0(x)y = g(x)$ . Here:

$$a_2(x) = (1-x), \quad a_1(x) = -4x, \quad a_0(x) = 5, \quad g(x) = \cos x$$

The dependent variable  $y$  and its derivatives  $y', y''$  appear solely to the first power with no products ( $yy'$ ) or nonlinear functions ( $\sin y, e^y$ ). All coefficients depend strictly on the independent variable  $x$ . Hence, the equation is **linear**.

OFFICIAL BOOK ANSWER: Second-order; linear.

Exact Match [Verified]

## Problem 1.1.3 (Zill 7th Ed Section 1.1 #3)

ORDER &amp; LINEARITY

**QUESTION:** State the order of the given ODE and determine whether it is linear or nonlinear:

$$t^5 y^{(4)} - t^3 y'' + 6y = 0$$

**Step 1 (Find Order):** The highest derivative present is the fourth derivative  $y^{(4)} = \frac{d^4y}{dt^4}$ . Thus, the order is **4**.

**Step 2 (Test Linearity):** Dependent variable is  $y$ , independent variable is  $t$ . The coefficients  $a_4(t) = t^5$ ,  $a_2(t) = -t^3$ ,  $a_0(t) = 6$  depend only on  $t$ .  $y$  and its derivatives are strictly linear. Hence, the equation is **linear**.

OFFICIAL BOOK ANSWER: Fourth-order; linear.

Exact Match [Verified]

## Problem 1.1.5 (7th Ed #5)

NONLINEAR ROOT

**QUESTION:**  $\frac{d^2y}{dx^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

**1. Order:** Highest derivative is  $\frac{d^2y}{dx^2} \implies$  **Order 2**.

**2. Linearity:** The first derivative  $\frac{dy}{dx}$  is squared and inside a square root operation. This violates linearity  $\implies$  **Nonlinear**.

BOOK ANSWER: Second-order; nonlinear (due to radical).

## Problem 1.1.6 (7th Ed #6 · 6th Ed #4)

NONLINEAR TRIG

**QUESTION:**  $\frac{d^2u}{dr^2} + \frac{du}{dr} + u = \cos(r+u)$

**1. Order:** Highest derivative is  $\frac{d^2u}{dr^2} \implies$  **Order 2**.

**2. Linearity:** Dependent variable  $u$  appears inside the nonlinear cosine function  $\cos(r+u) \implies$  **Nonlinear**.

BOOK ANSWER: Second-order; nonlinear (due to  $\cos(r+u)$ ).

## SECTION 1.1 (CONT.): PROBLEMS 8, 10, 11, 13

Linearity &amp; Explicit Verification

## Problem 1.1.8 (7th Ed #8 · 6th Ed #10)

RAYLEIGH EQ

**QUESTION:**  $\ddot{x} - (1 - \frac{1}{3}\dot{x}^2)\dot{x} + x = 0$ **1. Order:** Highest derivative is  $\ddot{x} = \frac{d^2x}{dt^2} \Rightarrow$  **Order 2.****2. Linearity:** Expanding gives  $\ddot{x} - \dot{x} + \frac{1}{3}\dot{x}^3 + x = 0$ . The cubic derivative term  $\dot{x}^3$  is nonlinear  $\Rightarrow$  **Nonlinear.****BOOK ANSWER:** Second-order; nonlinear (due to  $\dot{x}^2\dot{x}$ ).

## Problem 1.1.10 (7th Ed #10)

REVERSED ROLES

**QUESTION:**  $\frac{dx}{dy} + y^3x = \sin y$ **1. Order:** Highest derivative is  $\frac{dx}{dy} \Rightarrow$  **Order 1.****2. Linearity:** Dependent variable is  $x$ , independent variable is  $y$ . Fits  $x' + P(y)x = Q(y)$  with  $P(y) = y^3, Q(y) = \sin y \Rightarrow$  **Linear in  $x$ .****BOOK ANSWER:** First-order; linear in  $x$ .

## Problem 1.1.11 (Zill 7th Ed Section 1.1 #11 · 6th Ed #3)

DIFFERENTIAL FORM

**QUESTION:** Determine whether the given first-order differential equation is linear in the indicated dependent variable by matching with  $a_1(t)\frac{dy}{dt} + a_0(t)y = g(t)$ :

$$(y^2 - 1)dx + x dy = 0; \quad \text{(a) in } y; \quad \text{(b) in } x$$

**Part (a) Treating  $y$  as dependent:** Divide by  $dx$  to obtain standard form:

$$x \frac{dy}{dx} + y^2 = 1 \iff x \frac{dy}{dx} + y^2 - 1 = 0$$

The term  $y^2$  raises the dependent variable to power 2. Therefore, it is **nonlinear in  $y$ .****Part (b) Treating  $x$  as dependent:** Divide by  $dy$  to obtain standard form:

$$(y^2 - 1) \frac{dx}{dy} + x = 0$$

Here the dependent variable is  $x$ . Both  $\frac{dx}{dy}$  and  $x$  appear solely to the first power, and the coefficient  $(y^2 - 1)$  depends only on the independent variable  $y$ . Therefore, it is **linear in  $x$ .****OFFICIAL BOOK ANSWER:** Nonlinear in  $y$  because of  $y^2$ ; Linear in  $x$  with coefficient  $(y^2 - 1)$ .

Exact Match [Verified]

## Problem 1.1.13 (Zill 7th Ed Section 1.1 #13 · 6th Ed #11)

EXPLICIT VERIFICATION

**QUESTION:** Verify that the indicated function is an explicit solution of the given ODE. Assume an appropriate interval  $I$  of definition:

$$2y' + y = 0; \quad y = e^{-x/2}$$

**Step 1 (Compute Derivative):** Given  $y = e^{-x/2}$ , differentiate using the chain rule:

$$y' = \frac{d}{dx} [e^{-x/2}] = -\frac{1}{2}e^{-x/2}$$

**Step 2 (Substitute into LHS):**

$$\text{LHS} = 2y' + y = 2\left(-\frac{1}{2}e^{-x/2}\right) + e^{-x/2} = -e^{-x/2} + e^{-x/2} = 0 = \text{RHS}$$

Since the equality holds identically for all real numbers,  $y(x) = e^{-x/2}$  is an explicit solution on the interval  $I = (-\infty, \infty)$ .**OFFICIAL BOOK ANSWER:** From  $y = e^{-x/2}$ ,  $y' = -\frac{1}{2}e^{-x/2}$ . Then  $2y' + y = -e^{-x/2} + e^{-x/2} = 0$ . Interval:  $(-\infty, \infty)$ .

Exact Match [Verified]

## SECTION 1.1 (CONT.): PROBLEMS 14, 21, 23

Implicit Solutions &amp; Linear Systems

**Problem 1.1.14** (Zill 7th Ed Section 1.1 #14 · 6th Ed #12)

EXPLICIT VERIFICATION

**QUESTION:** Verify that the indicated function is an explicit solution of the given differential equation:

$$\frac{dy}{dt} + 20y = 24; \quad y = \frac{6}{5} - \frac{6}{5}e^{-20t}$$

**Step 1 (Compute Derivative):**

$$\frac{dy}{dt} = \frac{d}{dt} \left[ \frac{6}{5} - \frac{6}{5}e^{-20t} \right] = 0 - \frac{6}{5}(-20)e^{-20t} = 24e^{-20t}$$

**Step 2 (Substitute into LHS):**

$$\text{LHS} = \frac{dy}{dt} + 20y = 24e^{-20t} + 20 \left( \frac{6}{5} - \frac{6}{5}e^{-20t} \right) = 24e^{-20t} + 24 - 24e^{-20t} = 24 = \text{RHS}$$

The equation is satisfied identically for all  $t \in (-\infty, \infty)$ .**OFFICIAL BOOK ANSWER:**  $dy/dt = 24e^{-20t}$ ; then  $dy/dt + 20y = 24e^{-20t} + 24 - 24e^{-20t} = 24$ . Interval:  $(-\infty, \infty)$ .

Exact Match [Verified]

**Problem 1.1.21** (Zill 7th Ed Section 1.1 #21 · 6th Ed #15)

IMPLICIT VERIFICATION

**QUESTION:** Verify that the expression is an implicit solution of the ODE. Find an explicit solution and give its interval  $I$  of definition:

$$\frac{dX}{dt} = (X-1)(1-2X); \quad \ln \left( \frac{2X-1}{X-1} \right) = t$$

**Step 1 (Implicit Differentiation):** Differentiating  $\ln(2X-1) - \ln(X-1) = t$  with respect to  $t$ :

$$\begin{aligned} \frac{2}{2X-1} \frac{dX}{dt} - \frac{1}{X-1} \frac{dX}{dt} &= 1 \implies \left[ \frac{2(X-1) - (2X-1)}{(2X-1)(X-1)} \right] \frac{dX}{dt} = 1 \\ \frac{2X-2-2X+1}{(2X-1)(X-1)} \frac{dX}{dt} &= 1 \implies \frac{-1}{(2X-1)(X-1)} \frac{dX}{dt} = 1 \implies \frac{dX}{dt} = -(2X-1)(X-1) = (X-1)(1-2X) \end{aligned}$$

This verifies that the relation is an implicit solution!

**Step 2 (Solve for Explicit Solution  $X(t)$ ):** Exponentiate both sides:

$$\frac{2X-1}{X-1} = e^t \implies 2X-1 = Xe^t - e^t \implies X(2-e^t) = 1-e^t \implies X(t) = \frac{e^t-1}{e^t-2}$$

**Step 3 (Interval of Definition):** The denominator is zero when  $e^t - 2 = 0 \implies t = \ln 2$ . Since a solution interval  $I$  must be continuous, the maximal intervals of definition are  $I_1 = (-\infty, \ln 2)$  or  $I_2 = (\ln 2, \infty)$ .**OFFICIAL BOOK ANSWER:**  $dX/dt = (X-1)(1-2X)$ ; Explicit  $X(t) = \frac{e^t-1}{e^t-2}$ ; Interval  $I = (-\infty, \ln 2)$  or  $(\ln 2, \infty)$ .

Exact Match [Verified]

**Problem 1.1.23** (Zill 7th Ed Section 1.1 #49 · 6th Ed #23)

SYSTEM OF ODES

**QUESTION:** Verify that the indicated pair of functions is a solution of the system of differential equations on  $(-\infty, \infty)$ :

$$\frac{dx}{dt} = x + 3y, \quad \frac{dy}{dt} = 5x + 3y; \quad x = e^{-2t} + 3e^{6t}, \quad y = -e^{-2t} + 5e^{6t}$$

**Step 1 (Differentiate  $x(t)$  and test Eq 1):**

$$\begin{aligned} \frac{dx}{dt} &= -2e^{-2t} + 18e^{6t} \\ x + 3y &= (e^{-2t} + 3e^{6t}) + 3(-e^{-2t} + 5e^{6t}) = (1-3)e^{-2t} + (3+15)e^{6t} = -2e^{-2t} + 18e^{6t} \equiv \frac{dx}{dt} \quad \checkmark \end{aligned}$$

**Step 2 (Differentiate  $y(t)$  and test Eq 2):**

$$\begin{aligned} \frac{dy}{dt} &= 2e^{-2t} + 30e^{6t} \\ 5x + 3y &= 5(e^{-2t} + 3e^{6t}) + 3(-e^{-2t} + 5e^{6t}) = (5-3)e^{-2t} + (15+15)e^{6t} = 2e^{-2t} + 30e^{6t} \equiv \frac{dy}{dt} \quad \checkmark \end{aligned}$$

Both equations are identically satisfied for all  $t \in (-\infty, \infty)$ .**OFFICIAL BOOK ANSWER:**  $dx/dt = x + 3y$  and  $dy/dt = 5x + 3y$  verified for all  $t$  on  $(-\infty, \infty)$ .

Exact Match [Verified]

## SECTION 1.2: INITIAL-VALUE PROBLEMS (PROBLEMS 7, 9, 11, 12)

Parameter Families &amp; Initial Conditions

## Problem 1.2.7 (Zill 7th Ed Section 1.2 #7)

SECOND-ORDER IVP

**QUESTION:** The family  $x(t) = c_1 \cos t + c_2 \sin t$  is a two-parameter family of solutions of  $x'' + x = 0$ . Find a solution satisfying the initial conditions:

$$x(0) = -1, \quad x'(0) = 8$$

**Step 1 (Apply  $x(0) = -1$ ):**

$$x(0) = c_1 \cos(0) + c_2 \sin(0) = c_1(1) + c_2(0) = c_1 = -1$$

**Step 2 (Differentiate & Apply  $x'(0) = 8$ ):**

$$x'(t) = -c_1 \sin t + c_2 \cos t \implies x'(0) = -c_1 \sin(0) + c_2 \cos(0) = c_2 = 8$$

**Step 3 (Assemble Solution):** Substituting  $c_1 = -1$  and  $c_2 = 8$  yields  $x(t) = -\cos t + 8 \sin t$ .

**OFFICIAL BOOK ANSWER:**  $x(t) = -\cos t + 8 \sin t$

Exact Match [Verified]

## Problem 1.2.9 (Zill 7th Ed Section 1.2 #9)

TRIG SYSTEM IVP

**QUESTION:** Find a solution of  $x'' + x = 0$  using  $x(t) = c_1 \cos t + c_2 \sin t$  satisfying:

$$x(\pi/6) = \frac{1}{2}, \quad x'(\pi/6) = 0$$

**Step 1 (Set up Linear System for  $c_1, c_2$ ):**

$$x(\pi/6) = c_1 \cos(\pi/6) + c_2 \sin(\pi/6) = c_1 \frac{\sqrt{3}}{2} + c_2 \frac{1}{2} = \frac{1}{2} \implies \sqrt{3}c_1 + c_2 = 1$$

$$x'(\pi/6) = -c_1 \sin(\pi/6) + c_2 \cos(\pi/6) = -c_1 \frac{1}{2} + c_2 \frac{\sqrt{3}}{2} = 0 \implies -c_1 + \sqrt{3}c_2 = 0 \implies c_1 = \sqrt{3}c_2$$

**Step 2 (Solve System):**

$$\sqrt{3}(\sqrt{3}c_2) + c_2 = 1 \implies 3c_2 + c_2 = 1 \implies 4c_2 = 1 \implies c_2 = \frac{1}{4}$$

$$c_1 = \sqrt{3} \left( \frac{1}{4} \right) = \frac{\sqrt{3}}{4}$$

**OFFICIAL BOOK ANSWER:**  $x(t) = \frac{\sqrt{3}}{4} \cos t + \frac{1}{4} \sin t$

Exact Match [Verified]

## Problem 1.2.11 (7th Ed #11)

HYPERBOLIC IVP

**QUESTION:**  $y = c_1 e^x + c_2 e^{-x}$  for  $y'' - y = 0$ ;  $y(0) = 1, y'(0) = 2$ .

**1. System:**  $y(0) = c_1 + c_2 = 1$ .

$$y'(x) = c_1 e^x - c_2 e^{-x} \implies y'(0) = c_1 - c_2 = 2.$$

**2. Solve:** Adding:  $2c_1 = 3 \implies c_1 = \frac{3}{2}$ .

$$\text{Then } c_2 = 1 - \frac{3}{2} = -\frac{1}{2}.$$

**BOOK ANSWER:**  $y = \frac{3}{2}e^x - \frac{1}{2}e^{-x}$

## Problem 1.2.12 (7th Ed #12)

SINH SOLUTION

**QUESTION:**  $y = c_1 e^x + c_2 e^{-x}$  for  $y'' - y = 0$ ;  $y(1) = 0, y'(1) = 1$ .

**1. System:**  $c_1 e + c_2 e^{-1} = 0 \implies c_2 = -c_1 e^2$ .

$$c_1 e - c_2 e^{-1} = 1 \implies 2c_1 e = 1 \implies c_1 = \frac{1}{2e}.$$

**2. Assemble:**  $c_2 = -\frac{e}{2}$ .

$$y = \frac{1}{2}e^{x-1} - \frac{1}{2}e^{-(x-1)} = \sinh(x-1).$$

**BOOK ANSWER:**  $y = \sinh(x-1)$

## SECTION 1.2: GRAPHICAL SOLUTION MATCHING (PROBLEMS 35 &amp; 38)

Geometric Analysis of IVPs

**Context for Problems 35–38:** The graph of a member of a family of solutions of a second-order DE  $y'' = f(x, y, y')$  is given. Match each solution curve with at least one pair of the following initial conditions:

(a)  $y(1) = 1, y'(1) = -2$

(b)  $y(-1) = 0, y'(-1) = -4$

(c)  $y(1) = 1, y'(1) = 2$

(d)  $y(0) = -1, y'(0) = 2$

(e)  $y(0) = -1, y'(0) = 0$

(f)  $y(0) = -4, y'(0) = -2$

**Problem 1.2.35** (Zill 7th Ed Section 1.2 #35 · Figure 1.2.7)

CURVE MATCHING

**QUESTION:** Match the solution curve in **Figure 1.2.7** (a U-shaped curve entering from top-left, crossing  $x = -1$  at  $y = 0$  downwards, reaching a minimum at  $x \approx 0.5, y \approx -4.5$ , and rising steeply through  $(2, 0)$  towards  $(3, 5)$ ).

**Test 1 — Evaluate at  $x = -1$ :** Looking at the graph, at  $x = -1$ , the curve crosses the  $x$ -axis exactly at  $y = 0$ . Furthermore, the curve is decreasing rapidly (negative slope). Checking condition (b):  $y(-1) = 0$  and  $y'(-1) = -4 < 0$ . This is an exact physical match!

**Test 2 — Evaluate at  $x = 0$ :** At  $x = 0$ , the curve crosses the  $y$ -axis at  $y = -4$ . The curve has not yet reached its local minimum, so the tangent slope is still negative. Checking condition (f):  $y(0) = -4$  and  $y'(0) = -2 < 0$ . This is also an exact physical match!

**Conclusion:** The curve in Figure 1.2.7 simultaneously satisfies both initial conditions (b) and (f).

**OFFICIAL BOOK ANSWER:** Matches conditions (b) and (f).

Exact Match [Verified]

**Problem 1.2.38** (Zill 7th Ed Section 1.2 #38 · Figure 1.2.10)

CURVE MATCHING

**QUESTION:** Match the solution curve in **Figure 1.2.10** (an oscillatory wave curve with a peak at  $(0, 2.5)$ , descending through  $(1, 1)$ , crossing  $y = 0$  near  $x \approx 1.8$ , reaching a valley at  $x \approx 4, y \approx -4.5$ , and rising towards  $(7, 0)$ ).

**Test 1 — Evaluate candidate point  $x = 1$ :** The curve passes cleanly through  $(x, y) = (1, 1)$ . At this point, the curve is on its downward slope from the peak at  $(0, 2.5)$  towards the  $x$ -axis. Therefore, the tangent slope must be negative:  $y'(1) < 0$ .

**Test 2 — Compare candidate conditions at  $x = 1$ :**

- Condition (c) states  $y(1) = 1, y'(1) = 2 > 0$  (would require curve to be rising). False.
- Condition (a) states  $y(1) = 1, y'(1) = -2 < 0$  (matches decreasing curve passing through  $(1, 1)$ ). **True!**

**Test 3 — Check points at  $x = 0$  and  $x = -1$ :** At  $x = 0, y \approx 2.5$  (rules out (d) and (e) where  $y(0) = -1$ ). At  $x = -1, y < 0$  (rules out (b) where  $y(-1) = 0$ ).

**OFFICIAL BOOK ANSWER:** Matches condition (a).

Exact Match [Verified]

**EXAM STRATEGY NOTE:** In Concordia graphical IVP exam questions, always check both the **function value** (point  $(x_0, y_0)$ ) and the **sign of the slope** ( $\text{sign}(y'(x_0))$ ). A curve with horizontal tangent has  $y' = 0$ , rising has  $y' > 0$ , falling has  $y' < 0$ .

## SECTION 1.3: DIFFERENTIAL EQUATIONS AS MODELS (PROBLEMS 1, 2, 5, 9)

Rate Equations &amp; Physical Balance

## Problem 1.3.1 (Zill Section 1.3 #1)

## POPULATION

**QUESTION:** Determine DE for population  $P(t)$  with immigration at constant rate  $r > 0$ . What is the DE for emigration at rate  $r > 0$ ?

**1. Balance:**  $\frac{dP}{dt} = (\text{Growth rate}) \pm (\text{Migration})$ . Natural growth is proportional to current population:  $kP$ .

**2. Immigration:** Adds individuals  $\implies \frac{dP}{dt} = kP + r$ .

**3. Emigration:** Removes individuals  $\implies \frac{dP}{dt} = kP - r$ .

**BOOK ANSWER:**  $\frac{dP}{dt} = kP + r$ ;  $\frac{dP}{dt} = kP - r$

## Problem 1.3.2 (Zill Section 1.3 #2)

## BIRTH-DEATH

**QUESTION:** Model changing population with birth rate  $b = k_1P$  and death rate  $d = k_2P$ .

**1. Net Rate:** Rate of change equals rate in (births) minus rate out (deaths):

$$\frac{dP}{dt} = b - d = k_1P - k_2P$$

**2. Factor:**  $\frac{dP}{dt} = (k_1 - k_2)P = kP$  where  $k = k_1 - k_2$ .

**BOOK ANSWER:**  $\frac{dP}{dt} = k_1P - k_2P = (k_1 - k_2)P$

## Problem 1.3.5 (Zill 7th Ed Section 1.3 #5)

## NEWTON'S COOLING

**QUESTION:** A cup of coffee cools according to Newton's Law of Cooling  $\frac{dT}{dt} = k(T - T_m)$ . From the given graph showing temperature vs. time, estimate the initial temperature  $T_0$ , ambient medium temperature  $T_m$ , and cooling constant  $k$ .

**Step 1 (Estimate  $T_0$  and  $T_m$ ):** From the graph, at time  $t = 0$ , the curve begins at  $T(0) = T_0 \approx 180^\circ\text{F}$ . As  $t \rightarrow \infty$ , the temperature levels out asymptotically at  $T_m \approx 75^\circ\text{F}$  (room temperature).

**Step 2 (Estimate Cooling Constant  $k$ ):** We observe from the graph's slope that when the coffee reaches  $T = 85^\circ\text{F}$ , the rate of cooling is approximately  $\frac{dT}{dt} \approx -1^\circ\text{F}/\text{min}$ . Substituting into Newton's law:

$$\frac{dT}{dt} = k(T - T_m) \implies -1 \approx k(85 - 75) = 10k \implies k \approx -\frac{1}{10} = -0.1$$

**OFFICIAL BOOK ANSWER:**  $T_0 = 180^\circ$ ,  $T_m = 75^\circ$ ,  $k \approx \frac{-1}{85-75} = -0.1$

Exact Match [Verified]

## Problem 1.3.9 (Zill 7th Ed Section 1.3 #9)

## BRINE MIXTURE TANK

**QUESTION:** A mixing tank initially holds 300 gallons of water in which 50 lb of salt are dissolved. Pure water is pumped in at 3 gal/min, and the well-stirred solution is pumped out at the same rate. Determine the differential equation for the amount of salt  $A(t)$  in the tank at time  $t$ .

**Step 1 (Conservation of Mass):**  $\frac{dA}{dt} = R_{\text{in}} - R_{\text{out}}$ .

**Step 2 (Rate In):** Pure water contains 0 lb/gal of salt:  $R_{\text{in}} = (3 \text{ gal/min}) \times (0 \text{ lb/gal}) = 0 \text{ lb/min}$ .

**Step 3 (Rate Out):** Inflow equals outflow, so volume is constant  $V(t) = 300 \text{ gal}$ . The salt concentration at time  $t$  is  $C(t) = \frac{A(t)}{300} \text{ lb/gal}$ .

$$R_{\text{out}} = (3 \text{ gal/min}) \times \left( \frac{A}{300} \text{ lb/gal} \right) = \frac{3A}{300} = \frac{A}{100} \text{ lb/min}$$

**Step 4 (Assemble DE):**  $\frac{dA}{dt} = 0 - \frac{A}{100} = -\frac{A}{100}$ , with initial condition  $A(0) = 50 \text{ lb}$ .

**OFFICIAL BOOK ANSWER:**  $\frac{dA}{dt} = -\frac{A}{100}$ ,  $A(0) = 50$

Exact Match [Verified]

## SECTION 1.3 (CONT.): PROBLEMS 10, 13, 16

Brine, Torricelli Leaking &amp; Series Circuits

## Problem 1.3.10 (Zill 7th Ed Section 1.3 #10)

BRINE INFLOW

**QUESTION:** In Problem 9, suppose instead that brine containing 2 lb/gal of salt is pumped into the 300-gallon tank at 3 gal/min, and the well-stirred solution leaves at 3 gal/min. Determine the differential equation for  $A(t)$ .

## Step 1 (Calculate Rate In):

$$R_{\text{in}} = (3 \text{ gal/min}) \times (2 \text{ lb/gal}) = 6 \text{ lb/min}$$

## Step 2 (Calculate Rate Out): Volume is constant at 300 gal.

$$R_{\text{out}} = (3 \text{ gal/min}) \times \left( \frac{A(t)}{300} \text{ lb/gal} \right) = \frac{A}{100} \text{ lb/min}$$

## Step 3 (Assemble DE):

$$\frac{dA}{dt} = R_{\text{in}} - R_{\text{out}} = 6 - \frac{A}{100}$$

OFFICIAL BOOK ANSWER:  $\frac{dA}{dt} = 6 - \frac{A}{100}$ 

Exact Match [Verified]

## Problem 1.3.13 (Zill 7th Ed Section 1.3 #13 · 6th Ed #11)

TORRICELLI'S LAW

**QUESTION:** Water leaks through a circular hole of radius 2 in at the bottom of a cylindrical tank with base radius 10 ft. Using Torricelli's law with discharge coefficient  $c$  and  $g = 32 \text{ ft/s}^2$ , find the differential equation for the water height  $h(t)$ .

Step 1 (Torricelli General Formula):  $A_w(h) \frac{dh}{dt} = -cA_h \sqrt{2gh}$ .

## Step 2 (Cross-Sectional Areas):

$$A_w = \text{base area} = 10 \times 10 = 100 \text{ ft}^2 \quad (\text{or circular } \pi \cdot 10^2 = 100\pi)$$

$$A_h = \pi r^2 = \pi \left( \frac{2}{12} \text{ ft} \right)^2 = \pi \left( \frac{1}{6} \right)^2 = \frac{\pi}{36} \text{ ft}^2$$

Step 3 (Substitute with  $g = 32 \text{ ft/s}^2$ ):

$$\frac{dh}{dt} = -\frac{cA_h}{A_w} \sqrt{2(32)h} = -\frac{c(\pi/36)}{100} \sqrt{64h} = -\frac{c\pi \cdot 8}{3600} \sqrt{h} = -\frac{c\pi}{450} \sqrt{h}$$

OFFICIAL BOOK ANSWER:  $\frac{dh}{dt} = -\frac{c\pi}{450} \sqrt{h}$ 

Exact Match [Verified]

## Problem 1.3.16 (Zill 7th Ed Section 1.3 #16 · 6th Ed #13)

SERIES CIRCUITS

**QUESTION:** (a) A series circuit contains a resistor  $R$  and capacitor  $C$  with applied voltage  $E(t)$ . Find the DE for charge  $q(t)$ .  
 (b) For a series circuit containing an inductor  $L$  and resistor  $R$ , find the DE for current  $i(t)$ .

(a) **RC Circuit:** Kirchhoff's Voltage Law:  $V_R + V_C = E(t)$ . Since  $V_R = Ri = R \frac{dq}{dt}$  and  $V_C = \frac{1}{C}q$ :

$$R \frac{dq}{dt} + \frac{1}{C}q = E(t)$$

(b) **RL Circuit:** Kirchhoff's Voltage Law:  $V_L + V_R = E(t)$ . Since  $V_L = L \frac{di}{dt}$  and  $V_R = Ri$ :

$$L \frac{di}{dt} + Ri = E(t)$$

OFFICIAL BOOK ANSWER: **RC:**  $R \frac{dq}{dt} + \frac{1}{C}q = E(t)$ ; **RL:**  $L \frac{di}{dt} + Ri = E(t)$ 

Exact Match [Verified]

SECTION 1.3 (CONT.): PROBLEM 19 & CHAPTER 1 MODELING SUMMARY

Vibrations & Newton's Second Law

Problem 1.3.19 (Zill 7th Ed Section 1.3 #19 · 6th Ed #19)

SPRING-MASS DYNAMICS

**QUESTION:** After a mass  $m$  is attached to a vertical spring, it stretches the spring by an amount  $s$  and comes to rest in an equilibrium position. If the mass is set in motion with vertical displacement  $x(t)$ , formulate the second-order differential equation governing free, undamped motion using Newton's Second Law and Hooke's Law.

**Step 1 (Equilibrium State):** At equilibrium, gravity force downward balances spring restoring force upward:

$$F_{\text{net}} = mg - ks = 0 \implies mg = ks$$

**Step 2 (Dynamic Motion):** When displaced by distance  $x(t)$  downwards (positive direction), total spring stretch is  $s + x$ . Newton's second law:

$$m \frac{d^2 x}{dt^2} = F_{\text{net}} = mg - k(s + x) = mg - ks - kx$$

**Step 3 (Cancel Equilibrium Terms):** Since  $mg - ks = 0$ :

$$m \frac{d^2 x}{dt^2} = -kx \iff m \frac{d^2 x}{dt^2} + kx = 0$$

Dividing by  $m$  gives the standard simple harmonic oscillator ODE:  $\frac{d^2 x}{dt^2} + \omega^2 x = 0$  where  $\omega = \sqrt{k/m}$ .

OFFICIAL BOOK ANSWER:  $m \frac{d^2 x}{dt^2} + kx = 0$  or  $\frac{d^2 x}{dt^2} + \frac{k}{m} x = 0$

Exact Match [Verified]

MASTER SYNTHESIS: DIFFERENTIAL EQUATIONS MODELS ACROSS CHAPTER 1

| Physical System   | Governing Physical Law                                                 | Differential Equation Model                                                   | Order & Type     |
|-------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------|
| Population Growth | Malthusian Growth + Net Migration                                      | $\frac{dP}{dt} = kP \pm r$                                                    | 1ST ORDER LINEAR |
| Thermal Cooling   | Newton's Law of Cooling                                                | $\frac{dT}{dt} = k(T - T_m)$                                                  | 1ST ORDER LINEAR |
| Mixing Tanks      | Mass Conservation ( $\frac{dA}{dt} = R_{\text{in}} - R_{\text{out}}$ ) | $\frac{dA}{dt} + \frac{r_{\text{out}}}{V(t)} A = r_{\text{in}} C_{\text{in}}$ | 1ST ORDER LINEAR |
| Draining Tanks    | Torricelli's Law of Efflux ( $v = \sqrt{2gh}$ )                        | $A_w(h) \frac{dh}{dt} = -c A_h \sqrt{2gh}$                                    | 1ST ORDER NONLIN |
| Series Circuits   | Kirchhoff's Voltage Law ( $\sum V_k = E$ )                             | $L \frac{di}{dt} + Ri = E(t)$ (RL)                                            | 1ST ORDER LINEAR |
| Spring Vibrations | Newton's Second Law + Hooke's Law                                      | $m \frac{d^2 x}{dt^2} + kx = 0$                                               | 2ND ORDER LINEAR |